Hrithik Das

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Summary

Motivated IoT and Robotics Engineering graduate with a strong interest in machine learning and real-world problem solving. Experienced in academic and research projects and active in leadership and co-curricular activities, with strong teamwork and communication skills. Passionate about continuous learning and contributing to academic, professional and ambassadorial roles.

Education

University of Frontier Technology, Bangladesh 2022 – 2026
BSc.(Engg) in IoT & Robotics Engineering CGPA 3.04/4.00

St. Gregory's High School and College 2018 – 2020
Higher Secondary Certificate in Science GPA 4.67/5.00

Bangladesh Bank Adarsha High School 2018
Secondary School Certificate in Science GPA 5.00/5.00

Work Experience

Research Assistant, UFTB Aug 2026 – Present

– Collect and organize medical data from hospitals for research purposes. Ensure data accuracy and confidentiality throughout the collection process.

Volunteer, UNDP Sep 2025 – Present

– Engaged in youth-focused initiatives and community programs, supporting inclusive development and amplifying youth voices. Contributed to evidence-based activities aligned with the Sustainable Development Goals.

Volunteer & Campus Representative, Futurenation Sep 2025 – Present

– Actively contributed to youth leadership and community engagement initiatives, empowering young voices and supporting inclusive, youth-led projects for sustainable social impact.

Campus Ambassador, Workwave Jan 2025 – Present

– Promoting Workwave's initiatives on campus, engaging students, and facilitating events to increase awareness, participation, and community involvement.

Data Enumerator, Futurenation Programme (UNDP) Oct 2025

– Successfully conducted the Tertiary Students' Entrepreneurship Aspiration Survey, contributing to youth empowerment and supporting the growth of entrepreneurship initiatives in Bangladesh.

Projects

Agri-RAG: A Retrieval-Augmented Multimodal Framework for Context-Aware Crop Disease Advisory

A multimodal RAG-based system that identifies crop diseases and provides context-aware treatment recommendations using images and agricultural data.

A Context-Aware IoT and ML Framework for Deployment-Oriented Crop Recommendation in Bangladesh

Developed an intelligent system using IoT sensors and machine learning to provide context-aware crop recommendations, supporting sustainable agriculture and improving decision-making for farmers in Bangladesh.

Smart Diabetes Prediction System Using Machine Learning

Leveraged machine learning algorithms to predict diabetes risk, enabling early intervention and personalized health insights to support preventive healthcare.

ClimaVision: Explainable IoT and Machine Learning Framework for Microclimate Forecasting

Implemented an IoT-ML framework to forecast microclimate conditions, providing actionable environmental data to enhance agriculture, urban planning, and climate resilience.

Smart Automated Fire Fighting Robot

Engineered an autonomous fire-fighting robot using sensors and control systems to detect and extinguish fires, enhancing safety and emergency response efficiency.

Construction Safety Measures in Mega Projects in Bangladesh

Analyzed construction safety practices in mega infrastructure projects in Bangladesh using secondary data, identifying gaps between safety regulations and on-site implementation and highlighting key challenges in enforcement, training, and safety culture.

Luxury Brands Customer Perception: An explorative study among lower class people of Bangladesh.

Conducted an exploratory study on luxury brand perception among lower-income consumers in Bangladesh, analyzing awareness, aspirational attitudes, and purchase constraints to understand symbolic consumption and market influence.

ChakriHUT: Job Search and Employee Hiring Platform

Built a web platform enabling both job seekers to find opportunities and employers to hire talent, streamlining recruitment and improving access to employment.

Khida Lagse: Food Ordering and Delivery Management Software

Created a software platform for ordering and delivering food, managing restaurant operations, and ensuring timely delivery to customers, improving efficiency and convenience for both restaurants and users.

Insurance Management System

Designed a software system to manage insurance policies, claims, and customer records efficiently, streamlining operations and improving data accuracy for insurers and clients.

Library Management System

Developed a system to manage books, track borrowing and returns, and maintain member records efficiently, enhancing library operations and user experience.

Skills

Machine Learning: Supervised and unsupervised learning, classification and regression models, data preprocessing and feature engineering, model training and evaluation, predictive modeling, scikit-learn

Data Analytics: Data cleaning, data visualization, descriptive statistics, survey data analysis, Python, Excel

Internet of Things (IoT): Sensor integration, microcontroller programming, data acquisition, real-time monitoring, basic IoT architecture

Robotics: Robot design fundamentals, motor control, sensor-based navigation, microcontroller interfacing, basic automation

Soft Skills: Youth leadership, Communication & Teamwork, Public speaking, Community engagement

Co-curricular Activities

President, UFTB Sanatani Sangha Jan 2025 – Present

– Currently serving as President, previously Vice President, leading the student organization and organizing annual Saraswati Puja "Bani Archana" events to foster cultural engagement and active student participation.

Executive Member, UFTB STEAM Club Jul 2023 – Oct 2024

– Assisted in organizing the National STEAM Olympiad, supporting STEM and Arts workshops and youth engagement initiatives to promote innovation, creativity, and collaborative problem-solving.

Organizing Secretary, Gregorian Chess Club Oct 2018 – Nov 2019

– Coordinated multiple club tournaments and events, managed schedules, and facilitated member engagement to promote strategic thinking and teamwork among students.

Executive Member, Gregorian Science Club Oct 2018 – Nov 2019

– Assisted in organizing the annual science fest and Olympiad, one of the top college science events in Bangladesh, while supporting workshops and student led initiatives to foster scientific curiosity and collaboration.

Executive Member, BBAHS Football Club Feb 2016 – Jan 2018

– Supported club activities, coordinated football events, and promoted teamwork and sportsmanship among members while contributing to the growth of the school's football program.

Executive Member, BBAHS Chess Club Feb 2016 – Jan 2018

– Assisted in organizing chess events and competitions, promoted strategic thinking, and encouraged teamwork among club members.

Co-Organizer, Bangladesh Chess Federation Jan 2016 – Jan 2017

– Coordinated chess events and tournaments at the national level, facilitating smooth execution and promoting competitive play and strategic thinking among participants.

Achievements

National ICT Olympiad 2025

– Semifinalist, selected among the top 50 participants in Bangladesh, demonstrating strong ICT knowledge, problem-solving skills, and technical aptitude.

National STEAM Olympiad 2024

– Selected as a national finalist for innovative problem-solving and interdisciplinary STEAM skills.

Robotics & IoT Scholarship 2021

– Awarded a full scholarship based on academic merit and technical potential in robotics and IoT technologies at Creative IT Institute.

National Chess Tournament 2016

– Secured 38th place in the Bangladesh Chess Federation national tournament, demonstrating strategic thinking, competitive gameplay, and participation at the national level.

Divisional Chess Tournament 2016

– Achieved 4th place in the Dhaka Division, demonstrating advanced strategic thinking, competitive gameplay, and consistent performance in a regional-level chess tournament organized by the Bangladesh Chess Federation.

Bishwo Shahitto Kendro 2014

– Awarded "Shera Pathak", recognizing outstanding reading and literary comprehension skills in a national-level competition organized by Bishwo Shahitto Kendro.

Language Skills

Bangla (Native) · **English** (Professional working) · **Hindi** (Intermediate) · **Japanese** (Basic)

References

Suman Saha
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